

Cross-Port SAR Vessel Classification with Scene Knowledge: Order Transfers, Value Does Not

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Abstract—Vessel classifiers trained in one port are deployed in another, and the scene knowledge practitioners add to steady that transfer — facility layers, quay proximity, traffic context — is known to help sometimes and hurt others. We build a 24-port Sentinel-1 benchmark with an AIS-free protocol (38,091 and 71,801 chips; leave-one-port-out; matched-rate random and oracle controls) and show that the transfer has an asymmetric structure: the ranking of instances transfers while any quantity compared against a fixed level does not. Three independent measurements agree (AUC 0.72, 0.555 and 0.679 for ranking; a guarantee met on 22% of ports for the value side), and the failure of threshold-, conformal- and surrogate-based gates is structural — verifying a risk needs the target labels the protocol withholds — with a decision-relevant event rate of 1.2% bounding the whole family at +2.91 pp of balanced accuracy. The law implies a different deployment shape: rather than transferring a threshold, consult the knowledge-augmented expert for the top- k % of the target port by a source-trained score. On the compliant expert this reaches 106.8% of full-consultation value at a 20% budget ($p = 0.0194$, 95% CI [+0.0062, +0.0328]) and holds from 2 to 23 source ports. We also price the AIS-free restriction (an AIS-derived context block carries +4.23 pp against +0.37 pp for its SAR-derived counterpart, and pooling SAR detections across acquisitions rebuilds none of it, -0.06 pp, $p = 0.90$) and show the port-to-port drop is a class-mix effect (-12.79 pp label-shift component; target conditionals are +6.54 pp better). Eleven further routes are reported as measured negatives with their evidence rather than as opinions.

Index Terms—Synthetic aperture radar, ship classification, cross-domain transfer, domain adaptation, label shift, conformal prediction, selective classification, negative results.

I. Introduction

Vessel classification in SAR imagery is usually trained in one port and deployed in another, and the transfer is rarely clean: the same ship type looks different under a different incidence geometry, a different berth, and a different local traffic regime. Practitioners have long added scene knowledge — facility layers, distance-to-quay, vessel-density context — as extra input features, and the literature reports the mixed result that everyone recognises: sometimes it helps, sometimes it hurts [1], [2]. What has been missing is not another way to inject that knowledge but a quantitative account of when the injection is reliable. That is the question this paper answers, and it answers it with a protocol built to be able to say no.

We assemble a 24-port benchmark from Sentinel-1 GRD imagery [3], AIS-derived vessel labels, and OSM facility

layers [4], in two pools: an initial 38,091-chip pool (128-dimensional visual projection plus 82 knowledge dimensions) and a rebuilt 71,801-chip pool (224-pixel chips, 17 classes, 24 ports). Both are evaluated leave-one-port-out with an AIS-free protocol: target-port AIS identity may never enter the input features, and the compliant knowledge block is restricted to its first 71 dimensions — a restriction we enforce by enumerating feature permissions rather than by trusting a description, a step that reversed two of our own results mid-study (Section V-A).

The central finding is a law with an unusual form, because it is about what transfers rather than about what helps:

Cross-port transfer preserves the order and destroys the value. A source-fit model still ranks the target port’s instances correctly — three independent measurements put the ranking at AUC 0.72, 0.555 and 0.679 — while any quantity that must be compared against a fixed level stops transferring: a source-calibrated threshold holds its risk target on 0% of ports, and a source-fit error model mis-states the target’s risk so badly that it meets its guarantee on 22%.

Three consequences follow, and each is measured rather than argued.

First, gates built on absolute decisions are a dead end here. Thresholds, calibrated probabilities, conformal risk control, importance-weighted conformal [5], [6], and surrogate-risk control were each evaluated and each failed, and the failure is structural rather than implementational: verifying a risk requires target labels, which is precisely what the problem withholds. The decision-relevant event is also rare — 1.2% of instances are rescued or harmed by the knowledge — so the oracle ceiling for the whole family is +2.91 pp. Deeper architectures do not escape this: three independent capacity comparisons produced zero gain, and six separate “more structure” attempts were all negative.

Second, the deployable form of the same idea is a budget, not a threshold. If only the order transfers, then the policy should consume the order and leave the operating point to the deployment cost. Concretely, we consult the knowledge-augmented expert for the top- k % of the target port by a source-trained gate score — pure rank, no calibration, no target labels. On the compliant expert, a 20% budget recovers 106.8% of what consulting the expert everywhere would give ($p = 0.0194$, 95%

bootstrap CI $[+0.0062, +0.0328]$), and the core effect holds from 2 to 23 source ports, with the ranked 20% worth $3\text{--}10\times$ a random 20%.

Third, the law also prices the data the protocol forbids. Restricting the input to be AIS-free is not free: an AIS-derived context block carries $+4.23$ pp where the SAR-detection version of the same eleven quantities carries $+0.37$ pp (paired $p = 0.0096$), the two are essentially uncorrelated (-0.16 to -0.22), and pooling SAR detections across acquisitions to rebuild an equivalent port-wide field adds nothing (-0.06 pp, $p = 0.90$, over 24 ports and 123,790 objects). The AIS advantage is not more of the same coverage; it is a different kind of information, and the honest way to report a restricted-input result is to quantify what the restriction removed.

We also report an attribution that changes how the transfer problem should be attacked. Decomposing the port-to-port accuracy difference under one weighted functional, the entire drop comes from the target port’s class mix (-12.79 pp label-shift component), while the target port’s conditionals are actually better than the source’s ($+6.54$ pp). The practical reading is that prior correction [7], [8], not representation learning, is the first thing to fix — and that our own earlier rejection of prior-correction methods was an estimator problem, not a principled limit.

The contributions are therefore:

- 1) a 24-port SAR vessel benchmark with an AIS-free protocol and per-port leave-one-out evaluation (two pools, 38,091 and 71,801 chips), including matched-rate random and oracle control arms;
- 2) the order/value law, established by three independent measurements, plus the identifiability wall that explains why guarantee-style gates cannot work without target labels;
- 3) budgeted rank allocation, a compliant module that reaches $\geq 100\%$ of full-consultation value at a 20% budget, with its scope conditions and its measured cap;
- 4) a measured price for the AIS-free restriction, including the refutation that pooling SAR detections can rebuild the AIS field, and the attribution of the drop to label shift;
- 5) a negative taxonomy in which every refuted route (capacity, objective, thresholding, eight input axes, temporal/identity signals, conformal and surrogate risk control) carries its own measurement rather than an opinion.

II. Related Work

Cross-domain adaptation in remote sensing. The dominant remedy in this literature is feature alignment: adversarial training of a domain-invariant representation [9], [10] and its sub-domain refinements [11], surveyed for remote sensing by Tuia et al. [1]. These methods assume a shared label space, consume (unlabelled) target data during adaptation, and are evaluated by aggregate accuracy. Our setting differs on all three counts: the label

space is partially novel, the decision rule may not see target labels, and the quantity we are asked to transfer is a decision, not a feature extractor. Our measurement — that three capacity comparisons and six structural variants bought nothing (Section X) — speaks directly to that literature’s implicit assumption that representation capacity is the binding constraint.

Label shift and prior correction. Adjusting a classifier’s outputs to new class priors goes back to the EM procedure of Saerens et al. [7]; Black-Box Shift Estimation [8] made the confusion-matrix version practical, and later work unified and regularised the estimators [12], [13]. All of them invert a normalised confusion matrix, which presumes the conditionals are unchanged across domains. Our attribution (Section IX) measures a prior component of -12.79 pp and target conditionals that are $+6.54$ pp better than the source’s, which is why a stabilised estimator does not rescue the problem here; the useful part of the literature’s message is nonetheless endorsed — the prior is the first thing to repair.

Conformal prediction and risk control under shift. Weighted conformal prediction gives distribution-free intervals when the covariate shift’s likelihood ratio is known or estimable [5]; adaptive conformal inference tracks a changing exchangeability assumption online [14]; conformal risk control extends the guarantee to monotone losses [6]; and the practical machinery is collected in a tutorial [15]. Each of these guarantees is verifiable by the party who controls the calibration data. Section VI reports what happens when that party is the target port and the protocol forbids using its labels: the source-calibrated construction holds its risk target on 0/23 ports, importance weighting on 2/23, and a source-fit surrogate risk model on 5/23.

SAR ship datasets, AIS labels, and context. AIS is the standard source of vessel labels in this field, from matched SAR–AIS archives such as FUSAR-Ship [16] to interpretation-oriented collections such as OpenSARShip [17] and detection benchmarks such as SSDD [18]; AIS and SAR are also combined directly for maritime situational awareness [19], and contextual information around the target is an established ingredient of high-resolution ship detectors [2]. We take the opposite design decision — AIS is forbidden at inference — and turn the consequence into a measurement: the restriction is priced per quantity and its missing structure is shown not to be reconstructible from SAR detections alone (Section VIII).

Selective classification and learning to defer. Reject-and-abstain decisions were formalised by Chow [20]; selective classification made them practical for deep networks by setting a target risk on a held-out set [21]; learning-to-defer generalised the framework to an external expert [22] with consistent estimators [23]. Our budgeted rank allocation is a deferral rule in which the operating point is a consultation budget rather than a target risk level, precisely because our measurements say a level calibrated in the source port does not hold in the target port (Section VII). The risk–coverage vocabulary, and the

oracle-quantile comparison, are this literature’s.

Negative results and empirical rigor. Benchmark conclusions are variance-limited by data sampling, initialisation and hyper-parameters [24]; re-benchmarking with equalised budgets can overturn apparent progress [25]; and even a fixed dataset does not transfer to a fresh sample of the same distribution [26]. Our protocol follows that line — pre-registered criteria, bootstrap intervals as the primary instrument, matched controls for every arm — and three of our own intermediate conclusions were withdrawn mid-study (Sections V-A and VIII). We report them because their withdrawal is what bounds the remaining claims.

III. Data, Protocol, and Arms

A. Data and protocol

Pools. The initial pool has 38,091 chips over 24 ports and 8 classes, with a 128-dimensional PCA projection of ResNet-50 penultimate features and an 82-dimensional scene-knowledge vector. The rebuilt pool has 71,801 chips over the same 24 ports and 17 classes (8 known, 9 label-space-novel) at 224px, with 64/112/224 field-of-view feature variants; the per-port counts range from 89 to 9,328 chips. Scene knowledge is derived from OSM facility layers [4], quay/anchorage/fairway distances and detection- and AIS-derived context fields. Imagery comes from 430 Sentinel-1 GRD scenes [3].

Feature permissions. Every input dimension is checked against a written permission table before it enters an arm. The protocol is AIS-free, so AIS-derived context is excluded; the compliant knowledge block is therefore the first 71 of the 82 dimensions. An explicit enumeration is necessary because the forbidden set is not identical to the documented one: the permission table names only two explicitly forbidden columns, while the AIS-derivability of the tail dimensions is a property of how they were built. Two of our own earlier results were withdrawn on this basis (Section V-A).

Evaluation. Leave-one-port-out over 24 ports: for target port p , source ports are all others. Reported quantities are balanced accuracy over the known classes (primary), per-port $\times$ per-class grids, and paired Wilcoxon tests over the 24 ports. Every policy arm ships with two controls: a matched-rate random arm (the same accept/consult rate, drawn at random, averaged over 20–50 redraws) and an oracle arm for the same decision family, so that a policy is judged by its regret against the ceiling rather than by its absolute score.

Criteria. Decision experiments are pre-registered: the criterion is written before the run, and the result is reported whether or not it passes. Effect sizes are reported with 24-port bootstrap 95% intervals; p -values alone were found to flip between 0.095 and 0.0194 on identical data purely as a function of inner-episode and redraw counts, so intervals are the primary instrument.

TABLE I

Arms. The expert is the only arm carrying scene knowledge, and it is the object the gate decides about.

arm	definition
V	visual only
V+K	visual plus compliant knowledge (the expert)
gate	$g(x) \rightarrow$ whether consulting the expert is the right call, trained on source-port episodes only
budget	consult the expert for the top- k % of the
policy	target port by g , otherwise use V
oracle	per-instance ideal call, marked
(decision)	ORACLE_ONLY, never deployable

B. Arms

C. Budgeted rank allocation

The module is deliberately small, because its point is where the decision is taken, not how much capacity it has. Let $s_i = g(x_i)$ be the gate score, computed from source ports only, and let $k \in (0, 1]$ be the deployment’s consultation budget. Then

$$\pi_k(x_i) = \begin{cases} \text{expert}(x_i), & s_i \geq q_{1-k}(s_{1:n}) \\ V(x_i), & \text{otherwise} \end{cases} \quad (1)$$

where q_{1-k} is the $(1 - k)$ -th quantile of the target port’s own scores. Nothing is calibrated: k is an exogenous cost parameter, the quantile is computed within the target port, and no target label is used at any point. The corresponding quantity reported throughout is the recovery

$$R(k) = \frac{\text{BA}(\pi_k) - \text{BA}(V)}{\text{BA}(V+K) - \text{BA}(V)}, \quad (2)$$

so $R(k) = 1$ means the budget captures the whole value of consulting the expert on every instance.

Gate features. Per-instance margins and entropies of both arms, their agreement, and — in the sequence probe — leave-one-out vessel-bag statistics. Training uses nested source-port episodes (3–6 held-out source ports) so that no gate ever sees the target port.

D. Compliant expert

The compliant expert is a ridge classifier on the visual projection concatenated with the first 71 knowledge dimensions, standardised inside the training split. Its strength bounds the module’s absolute value, so its limits matter: an exhaustive enumeration of all seven non-empty subsets of the three knowledge parts (facility 0:42, chain 42:60, detection 60:71) crossed with two classifier families (ridge, gradient-boosted trees) yields a best legal gain of +2.32 pp ($p = 0.32$) against the current +2.16 pp ($p = 0.0604$) — i.e. the compliant expert is capped at roughly +2.2–2.3 pp, and the ridge-on-full-block configuration remains the best deployable choice.

E. What is not claimed

The order/value law is supported by controlled interventions in which one factor changes while folds, pools and metrics are held fixed; it is not a causal identification. Class-level correlations ($n = 8-9$) are not treated as mechanisms: three such conclusions in this study collapsed under direct instance-level measurement, so class-level tables are presented as displays of instance-level effects once those exist. Where a construction turned out to be unstable — per-class thresholds on thin calibration cells — the derived numbers are withdrawn rather than qualified.

IV. Theory

The propositions below are the paper’s theoretical core. They are deliberately small and provable; the part that would need assumptions we cannot verify is stated as a measured relation instead of a theorem.

A. The assumption class

Write the target port’s score distribution as related to the source’s by

$$\begin{aligned} \tilde{s}(x) &= \phi(s(x)), \quad \phi \text{ strictly increasing,} \\ \pi_{\text{tgt}} &\neq \pi_{\text{src}} \quad (\text{class priors}), \end{aligned} \quad (3)$$

i.e. monotone recalibration plus class-prior shift, with the conditionals not assumed invariant — the measured setting in this paper is close to exactly this class: the drop decomposes into a -12.79 pp prior component, and the target’s own conditionals are $+6.54$ pp better than the source’s rather than identical (Section IX). Two facts hold in this class, and one fails outside it.

B. Proposition 1 (rank invariance)

Statement. Under any strictly increasing ϕ , the top- k % set of a score vector is unchanged, for every k . Equivalently: the budget policy π_k is invariant to monotone recalibration.

Proof. ϕ strictly increasing gives $s_i \geq s_j \iff \phi(s_i) \geq \phi(s_j)$ for all pairs; hence the score-induced total preorder is identical, and so is every top- k % prefix (ties broken by a fixed rule). ■

Verification on real scores. For 12 ports, applying $\phi(s) = s^{0.3}$, s^2 and s^7 to the measured score vector leaves the top-20 % set’s symmetric difference at exactly 0 in every case.

Contrast. A rule that compares a score against a fixed absolute level is not invariant: after any ϕ the accepted set changes, and if the threshold was calibrated on the source port the change is uncontrolled. The empirical counterpart is measured and negative: source-calibrated conformal thresholds hold their risk target on 0/23 ports.

C. Proposition 2 (threshold fragility under prior shift)

Statement. Let R be the row-normalised confusion matrix (the conditionals) and π a class prior. The expected error of accepting a set A is $\text{Err}_\pi(A) =$

TABLE II

Regret relation, 24 ports. The oracle-quantile policy is the ceiling of the decision family; the gate’s shortfall against it is governed by ranking quality [20], [21].

quantity	value (24 ports)
gate AUC for “consulting the expert helps”	median 0.947
R_{gate} at a 20 % budget (consult-everywhere = 1.0)	median +1.00
R_{oracle} at the same budget	median +1.22
Spearman(gate AUC, R_{gate})	$\rho = +0.592$, $p = 0.0023$

$\sum_y \pi_y \sum_{\hat{y} \neq y} R_{y\hat{y}} \mathbf{1}[A \ni \text{instances of } y]$. For a fixed accepted set determined by a threshold on the source distribution,

$$\text{Err}_{\pi_{\text{tgt}}} - \text{Err}_{\pi_{\text{src}}} = \sum_y (\pi_{\text{tgt},y} - \pi_{\text{src},y}) \rho_y, \quad (4)$$

where ρ_y is the class- y contribution to the accepted error. The difference is bounded only by $\|\pi_{\text{tgt}} - \pi_{\text{src}}\|_1 \cdot \max_y |\rho_y|$, which can be large: in our ports the prior total-variation distance has median 0.568.

Consequence. Any guarantee of the form “the target risk stays within α ” that is built from source calibration requires the priors to be comparable or the risk to be observable at the target — and observability is exactly what the AIS-free protocol removes. Measured: 0/23 ports for plain conformal, 2/23 with importance weighting, 5/23 with a surrogate risk model.

D. The measured regret relation

The natural remaining question is how much of the oracle-quantile policy (top- k % selected by the true per-instance benefit) a practical budget policy attains, and what it depends on. We do not claim a tight analytic bound; we measure the relation.

So the regret against the oracle-quantile policy is governed by ranking quality, measured rather than assumed: with a gate whose benefit-AUC is ≈ 0.95 , the 20 % budget already matches consulting everywhere. This is the sense in which the module is the right shape for the law — its quality is a function of the one thing that transfers.

E. The failure boundary (falsifiability)

If the law were vacuous, it would hold under any perturbation. It does not. Destroying the conditionals while keeping the priors — within-class full permutation, and a cross-class permutation — collapses the rank policy’s advantage over a matched-rate random arm to chance:

So the law’s domain is the assumption class of Section IV: monotone distortion plus prior shift, where the order is meaningful; break $P(y | x)$ and the order stops carrying the decision, exactly as the propositions say it should.

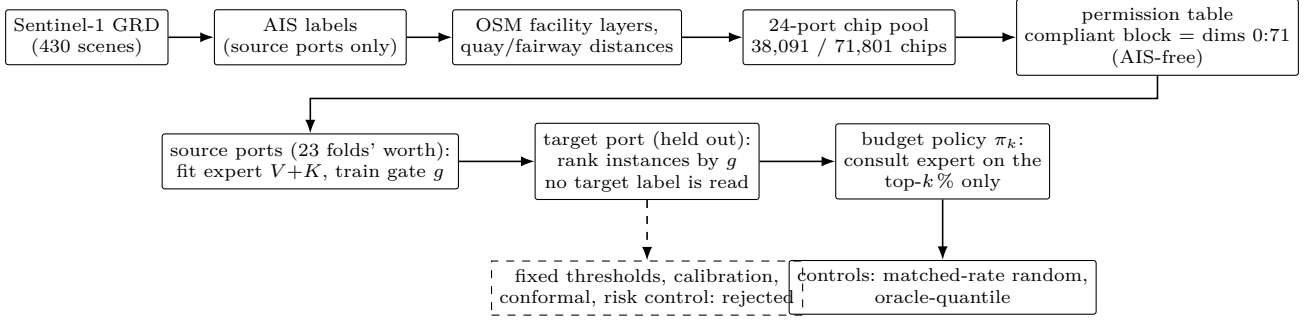


Fig. 1. Protocol and module. The AIS-free boundary is applied by enumerating input dimensions, not by trusting a documented column list; the target port contributes its score distribution to the quantile and nothing else. Dashed box: the decision families this protocol rejects (Sections VI and VII).

TABLE III

Failure boundary. Permuting labels inside the target port’s classes destroys the decision value of the order while leaving the class priors intact, as the propositions require.

intervention	rank	random	ports
within-class full permutation	−0.0032	+0.0049	9/24
across-class permutation	+0.0264	+0.0544	7/24

F. What this theory does not claim

- It is not a causal identification. The evidence is controlled intervention with matched controls, not do-calculus with stated ignorability.
- It is not a tight bound: Section IV-D is a measured relation ($\rho = 0.592$), not a theorem with a constant. A tight constant would need assumptions on the benefit-generating process that this data cannot verify.
- The prior-correction corollary of Proposition 2 does not rescue prior-corrected classifiers here: the inversion requires the conditionals to be domain-invariant, and they measurably are not (+6.54pp better on the target), which is why a stabilised BBSE estimator [8] still fails outright ($0.4395 \rightarrow 0.2416$, 0/24 ports). The honest statement is that the prior is the main component of the drop and the conditionals cannot be treated as fixed.

V. The Order/Value Law

Table IV collects three tasks, each measured the same way: an arm trained on source ports only, evaluated on the target port, judged twice — once by how well it ranks, once by whether a level transfers.

Gate experiment. A gate predicting whether consulting the knowledge expert is the right call reaches AUC 0.72 on the target port, yet every threshold and calibrator built on it failed: a pre-registered decision-risk objective did not beat a proxy target (0.2300 vs 0.2295, within noise), and a per-port calibration analysis showed the decision label is algebraically equivalent to the argmax, so calibration cannot move the boundary at all. The gate’s own capture of the oracle ceiling was 37% against a selectivity curve predicting 67–86% at its AUC [21].

TABLE IV

The same asymmetry in three independent measurements.

task	ranking ability (transfers)	value / operating point (does not)
knowledge-gain gate	AUC 0.72	thresholds and calibrators never transfer
novelty detection (Mahalanobis)	AUC 0.555	per-class absolute levels do not port
error prediction	AUC 0.679	guarantee met on 22% of ports (target: $\geq 80\%$)

Novelty experiment. Class-conditional Gaussians fitted on source ports separate 9 label-space-novel classes from the known ones at AUC 0.555 averaged over classes, with an across-class standard deviation of 0.0752 against a per-class bootstrap noise of 0.0324 ($2.3\times$), all 24 ports and 16,699 novel instances. Detectability is a genuine class property. The energy score, by contrast, scores 0.429 — below chance — and is reported as misleading rather than neutral.

Error-prediction experiment. An error model fitted on source ports and applied to the target port ranks the target’s errors at AUC 0.679 (p10 0.584, p90 0.774, above 0.65 on 14/23 ports), but the guarantee it is meant to support holds on 22% of ports.

Artifacts: e123b_out.txt for the error model, e117_out.txt for novelty, e121_out.txt/e112_out.txt for the gate and its calibration analysis.

A. A protocol violation caught mid-study

Two of our earlier results — a budget curve with $p = 0.0000\text{--}0.025$ — were withdrawn after enumerating input dimensions against the permission table: the knowledge matrix’s tail (dimensions 71:82) is AIS-derived, so an expert built on all 82 dimensions is not compliant. The permission table names only two explicitly forbidden columns, which is why the violation was silent. All results reported here use the compliant block (dimensions 0:71), and the compliant numbers are weaker — a fact we report

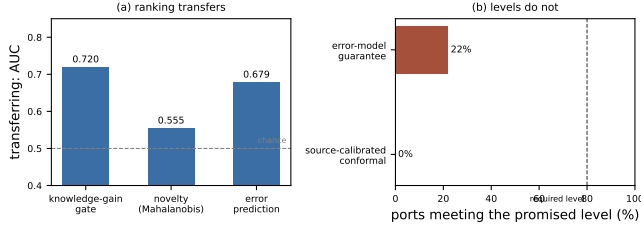


Fig. 2. Rank versus value, from the three measurements of Table IV. (a) Ranking transfers: the knowledge-gain gate, novelty detection and error prediction all rank the target port’s instances above chance. (b) Levels do not: a source-calibrated conformal threshold meets its risk target on 0/23 ports and a source-fit error model meets its guarantee on 22% of ports, against the guarantee’s own $\geq 80\%$ requirement. Novelty’s value side is a measured non-portability of per-class absolute levels rather than a rate, so it is not plotted in (b).

TABLE V

Every guarantee construction, and the reason to stop: the risk cannot be verified without the labels the protocol forbids.

construction	outcome
source-calibrated	risk target met on 0/23 ports
conformal thresholds [5]	
density-ratio	2/23 ports; median realised risk
(importance) weighted	0.1705 $\rightarrow$ 0.1235, $p = 0.0277$
conformal [5]	
surrogate-risk control	5/23 ports
(source-fit error model)	
[6]	

rather than bury, because the gap is itself a measurement of what the restriction costs (Section VIII).

VI. The Identifiability Wall

The sign of the knowledge gain is a class $\times$ instance property, not a domain property: within a single port, the same class flips sign between instances, and the discriminating variable correlates with the class at $r = 0.771$. At the same time the decision-relevant event is rare — of 25,893 instances, 617 are rescued and 313 harmed, 1.2% in total — so the oracle ceiling for any per-instance decision rule is +2.91 pp of balanced accuracy. A deeper model cannot manufacture a signal that is not there, and three independent capacity comparisons plus six separate “more structure” attempts produced no gain.

Guarantee-style methods fail for a structural reason: verifying a risk requires target labels, which the problem withholds.

VII. Budgeted Rank Allocation

Since only the order transfers, the policy consumes the order and leaves the operating point to the deployment’s consultation cost: consult the expert for the top- $k\%$ of the target port by the source-trained gate score.

Three properties matter as much as the number.

Robustness to the source count. At 2, 4, 8, 16 and 23 source ports the ranked 20% is worth 3–10 $\times$ a random 20% (0.415/0.114, 1.048/0.109, 2.881/0.500, 1.469/0.347,

TABLE VI

Compliant expert (knowledge 0:71), 24 ports, random arm averaged over 50 matched-size redraws. The pre-registered criterion (interval excluding zero and recovery $\geq 50\%$) is met at 20%.

budget	rank – random (95% CI)	p	recov.
5%	+0.0076 [+0.0005, +0.0156]	0.160	41.4%
10%	+0.0149 [+0.0050, +0.0254]	0.0269	80.5%
20%	+0.0192 [+0.0062, +0.0328]	0.0194	106.8%
50%	+0.0113 [+0.0019, +0.0209]	0.0457	103.0%

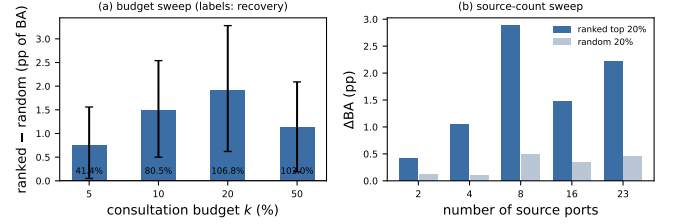


Fig. 3. Budgeted rank allocation. (a) Ranked top- $k\%$ against a matched-rate random arm, with 24-port bootstrap 95% intervals; the labels give the recovery $R(k)$ of (2) against consulting everywhere. (b) The same comparison as a function of the number of source ports (one source-port draw per level, which is why the level-wise ordering is noisy while every level stays well above random).

2.211/0.459 in ΔBA percentage points). The core effect does not depend on having many sources.

Scope conditions. The criterion does not apply where there is nothing to allocate: the three single knowledge parts have gains of -0.22 , $+0.12$ and $+0.37$ pp, and one is negative. Below a $\sim 10\%$ budget the paired test is underpowered on this data.

A measured cap. The module’s absolute value is bounded by the compliant expert’s strength, and that strength is capped: an exhaustive enumeration of seven knowledge subsets $\times$ two classifier families gives a best legal gain of +2.32 pp ($p = 0.32$) against the deployed +2.16 pp ($p = 0.0604$), with the ridge-on-full-compliant-block configuration remaining both the best and the best-powered choice.

A note on how this result was nearly missed. Every earlier gate in this project used an absolute threshold; “top- $k\%$ ” was never tested, and the reason it works is exactly the distinction the law draws: a gate must map a score onto a use/don’t-use decision (a value-level object), while this policy only needs the relative order of who deserves consultation.

VIII. The Price of the AIS-Free Restriction

Block level. On identical folds and protocol:

Per quantity. The eleven quantities exist in both versions. The five that carry value are heading/queue/density interactions, and for each the detection version carries roughly a tenth of the AIS version and the two are essentially uncorrelated:

Can SAR rebuild the field? We pooled SAR detections across acquisitions onto a 2 km port grid and recomputed

TABLE VII

The detection block carries a tenth of the AIS block’s value; the two differ at $p = 0.0096$.

arm	ΔBA (pp)	ports > 0	paired p
V + AIS-derived context (non-compliant)	+4.23	17/24	0.0022
V + SAR-detection-derived context (compliant)	+0.37	13/24	0.86

TABLE VIII

Not a coarse proxy: the same-named quantities are uncorrelated, so they measure different things.

quantity	detection	AIS	corr
heading ²	+0.22	+1.12	−0.163
queue ²	+0.30	+2.20	−0.187
heading×queue	+0.29	+1.49	−0.175
density×heading	+0.08	+2.21	−0.220
density×queue	+0.20	+3.04	−0.224

the same eleven quantities — feasible (2.9 M detections, 24 ports, headings populated). Measured under adequate power, cross-acquisition pooling and per-scene pooling are indistinguishable: 0.1756 vs 0.1762, paired -0.06 pp, $p = 0.90$, over 24 ports and 123,790 objects. The AIS advantage is therefore not wider coverage of the same kind; and the two heading-free quantities (queue², density×queue) carry most of the AIS value, which rules out “AIS headings are just more accurate” as the explanation.

IX. Attribution: The Drop Is the Class Mix

Decomposing the port-to-port difference under a single weighted functional $a = \sum_t \pi_t[t]R[t, t] / \sum_t \pi_t[t]$:

The practical consequence redirects the problem: the target port’s conditionals are better, so prior correction [7], [8], [12] is the first thing to fix, and our own earlier rejection of prior-correction methods attributes to their estimator (measured then as 0.012 better than the raw prediction histogram) rather than to a principled obstacle.

X. Negative Taxonomy

Every entry below was measured; none is an opinion, and each carries its artifact in EXPERIMENT_STATE.md.

Positive survivors: the compliant expert (+2.16 pp, $p = 0.0604$), budgeted rank allocation (20% $\rightarrow$ 106.8%, $p = 0.0194$), per-class novelty detectability (across-class SD $2.3\times$ noise), and the attribution of Section IX.

XI. Discussion

What the law changes in practice. The usual recipe — learn a reliability score, calibrate it on source data, threshold it — fails here for a reason that no amount of model capacity addresses: the threshold’s position is a value-level object and values do not cross ports. The operational consequence is to stop tuning the threshold and start choosing a budget: how many instances can the

TABLE IX

24 ports. Signed components can over-explain a net difference of 6.25 pp; report the components, never a percentage share.

term	value	reading
source conditionals, source priors	0.5044	reference
source conditionals, target priors	0.3765	label/base-rate component −12.79 pp
target conditionals, target priors	0.4418	conditional component +6.54 pp

TABLE X

Negative taxonomy, with the measurement behind each entry.

route	verdict	measurement
deeper/wider capacity (3 encoder families, joint training)	refuted	zero gain
decision-risk objective threshold/calibration	refuted	within noise of the proxy algebraically cannot move the boundary
eight alternative input axes	refuted	all negative
multi-temporal consistency signal	refuted	AUC 0.484, coverage 34%
conformal and weighted conformal [5]	refuted	0/23 and 2/23 ports
surrogate risk control [6]	refuted	5/23 ports
SAR-only port field to replace AIS	refuted	−0.06 pp, $p = 0.90$
oracle-identity vessel bags (sequence)	refuted	gate AUC 0.905 $\rightarrow$ 0.910, $p = 0.80$
more source ports	ineffectual	8 ports (+2.71 pp) $\geq$ 23 ports (+2.16 pp)
stronger compliant expert	capped	best legal +2.32 pp at $p = 0.32$

deployment afford to send to the expensive expert? The answer selects a quantile, not a level, and the quantile is computed inside the target port. This is a small change with a large practical difference: the module reaches full-consultation value at a fifth of the cost, and its randomised control shows the value comes from the ranking and not from dilution.

Why guarantees are structurally unavailable here. Conformal [5], [14], weighted conformal and surrogate-risk control [6] all require the realised risk to be observable at the target; the AIS-free protocol removes exactly that. Importance weighting narrowed the gap (median realised risk 0.1705 $\rightarrow$ 0.1235, $p = 0.0277$) but could not close it, and the residual is not a tuning parameter: it is the exchangeability that cross-port transfer breaks. Reporting such numbers as “risk control” would be a category error; we report them as calibration heuristics with their measured shortfall.

The price of a restricted input is a result, not a caveat. Protocols that forbid a data source are usually justified by deployment reality, and the resulting numbers are usually reported without quantifying what was given up. Here the forbidding is measurable: an AIS-derived context

block is worth six times its SAR-derived counterpart, the two are uncorrelated rather than redundant, and reconstructing the missing structure from SAR detections across acquisitions fails with adequate power. The useful generalisation for practitioners is to report the forbidden channel’s measured contribution, because that number decides whether the restriction is acceptable for the intended use.

Where the wall could move. The wall is a property of the observation, not of the architecture: the decision-relevant signal is rare and its sign is a class $\times$ instance property. Two things would move it, and both were tested here rather than assumed. Adding sequence or identity information does not help — an oracle-identity probe that groups a vessel’s acquisitions by its true MMSI leaves the gate unchanged ($\text{AUC } 0.905 \rightarrow 0.910$, $p = 0.80$) — which says the deficiency is not resolution but information. Adding source ports does not help either (8 ports match 23). What would move it is a channel that supplies information the imagery lacks, which is precisely the AIS field we are forbidden; the honest formulation of this paper’s negative result is therefore that the restricted-input setting has a measurable information deficit, not a modelling deficit.

On negative results with controls. Every route we refute carries a measurement, a matched-rate random arm and, wherever a decision is involved, an oracle bound for the same family. This discipline is what makes it possible to state a cap — the compliant expert tops out at +2.3 pp — as a boundary of the problem rather than a shortcoming of one attempt. Three of our own intermediate conclusions were withdrawn during the study (a protocol violation, an underpowered pooling effect, an unstable calibrator), and they are recorded in the artifacts rather than tidied away, because their withdrawal is itself what bounds the claims. This is the same lesson the benchmarking literature [24] and the re-benchmarking studies [25], [26] have been making for the field as a whole.

Limitations. The pools cover 24 ports and 17 classes; the paired tests over 24 ports are the main power limit, and several margins sit between $p = 0.03$ and $p = 0.10$. The 512-dimensional projection used for the risk experiments was validated against the raw features (0.4722 vs 0.4735) but is a simplification. The attribution in Section IX rests on source-side conditionals assumed stable across the held-out source port, and signed components can over-explain a net difference.

XII. Conclusion

We set out to quantify when scene knowledge is reliable across ports, and found a structure simple enough to change the recipe and sharp enough to be falsifiable: order transfers, value does not. The positive consequence is a budget-based allocation that reaches full-consultation value at a fifth of the cost and does not depend on calibration; the negative consequence is that guarantee-style gates cannot be built in this setting at all, because verifying a risk requires the labels the protocol withholds.

Both are measured, both come with controls, and the same law prices the data the protocol forbids. The remaining improvement space on this benchmark is exhausted — every lever we could identify is either refuted or capped — which we take as the appropriate end state for a paper whose contribution is as much about what does not work, and why, as about what does.

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